Topcugil and Hiziroglu
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REVIEW

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Neuromarketing and artificial intelligence: innovative approaches to customer experience management from a firm perspective

Beyza Topcugil[1*] and Abdulkadir Hiziroglu[1]

Abstract

The integration of advanced technologies has transformed marketing practices, making the roles of artificial intelligence and neuromarketing increasingly relevant to customer experience management. The aim of this study is to systematically conceptualize the interaction between neuromarketing, artificial intelligence, and customer experience management by developing a conceptual framework. Methodologically, the research employs a conceptual design supported by illustrative secondary vignettes, drawing on existing customer experience management and artificial intelligence models from the literature and extending them with neuromarketing applications. The framework suggests that neuromarketing data (e.g., EEG, eye-tracking, GSR) processed by artificial intelligence-driven analytics generate emotional, cognitive, and behavioral insights that enhance firms' ability to monitor, adapt, and design touchpoints across the customer journey. Illustrative vignettes drawn from publicly available secondary sources on three firms (TUI, Zyro, and Liberty Group) operating in distinct sectors show how such integration is associated with interface optimization, advanced recommendation systems, and platform enhancements, suggesting the potential for greater personalization, customer satisfaction, and firm efficiency. Beyond big data, neuromarketing offers unique consumer-level insights that broaden the scope of customer experience management, enabling businesses to transform B2C strategies through more precise segmentation and highly tailored interactions. However, the study also underscores ethical dilemmas such as transparency, consumer manipulation, and data privacy, which require careful management. Taken together, the proposed framework suggests that artificial intelligence–neuromarketing integration may function not only as a supportive tool but also as a transformative driver in customer experience management. By balancing technological innovation with ethical responsibility, firms can employ these technologies to pursue sustainable, trust-based value creation for both customers and businesses.

Keywords Artificial intelligence, Neuromarketing, Customer experience, Customer experience management, Illustrative vignettes

Introduction

The advent of emerging technologies that autonomously collect, process, and utilize data has simultaneously simplified daily life and raised consumer expectations. As consumers become more tech-savvy, they demand fast, seamless digital experiences and instant solutions [17]. Two key components of these solutions are customer experience management and artificial intelligence applications.

Customer Experience Management (CXM) involves understanding customer data at various touchpoints throughout the purchasing process, identifying cues perceived by customers, and sending appropriate signals to create value for them [11, 15, 21, 23]. In this article, we use the abbreviation CXM in alignment with the essential

*Correspondence:
Beyza Topcugil
beyza.topcugil@bakircay.edu.tr
[1] İzmir Bakırçay University, Izmir, Turkey

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CXM-focused research [11, 38] that employs this acronym to emphasize the concept's holistic, managerial, and experiential orientation. Artificial intelligence (AI) can generate insights in seconds that would otherwise be time-consuming and costly for humans [16, 25], while also enabling real-time data acquisition [1, 37]. These capabilities can give businesses a significant CXM advantage, setting them apart and offering a competitive edge. Recent syntheses confirm that the AI–CXM interface is one of the most rapidly evolving areas of marketing scholarship [19, 28], and at the same time the convergence of neuromarketing with AI has become a distinct research frontier [1, 7].

A review of the extant literature reveals a consensus that AI applications are significant for CXM, with ongoing discourse in this area. These discourses have led to diversification in the technology used by AI. In particular, studies on big data [11, 18], which is considered the raw material of AI, have become increasingly prominent [17]. However, what other data can AI learn from and offer unique advantages? When we ask this question, we come across a subject that has been recognized and applied in the sector for some time but has not been emphasized much in literature: neuromarketing. Developments in neuromarketing, the application of neuroscience to marketing, have provided insights into the complex inner world of the consumer. The reliance of neuromarketing applications on small-scale laboratory experiments has limited the applicability of the data obtained in this field to marketing practice. However, today, thanks to AI, individual data from the human brain can be used to train machines and transformed into vast bodies of information. AI and neuromarketing applications can be used together, and artificial intelligence algorithms fed with neuromarketing data can provide unique insights and responses for CXM. The collaboration between neuroscience and AI has gained conceptual momentum in areas such as the health sciences. Nevertheless, while neuromarketing and AI are innovative concepts discussed separately in the marketing literature [7, 28], there are very few studies addressing them jointly. Despite this gap, unique neurophysiological data derived from individuals are already being used by some companies in collaboration with artificial intelligence. Furthermore, leading scholars in CXM have proposed research questions on the contributions of neuroscience and technology to understanding customer experiences [21, 36], yet these questions remain largely unanswered in the literature. The marketing literature often lags behind practical developments, therefore, it is essential to bridge this gap by examining the practical implications and validity of such advancements [36]. We develop a conceptual framework that specifies how AI and neuromarketing inputs can be integrated into CXM workflows and illustrate the framework with secondary-case vignettes. Through this framework, we seek to both inspire practitioners with innovative and proactive ideas and enrich the academic literature that has thus far lagged behind practice.

Theoretical basis

Customer experience management refers to the management of all experiences that customers have with a product or business [28, 29]. These experiences encompass the entire customer journey, covering the pre-purchase, purchase, and post-purchase stages. CXM involves systematically identifying and prioritizing customer cues obtained throughout the customer journey with analytical tools, designing and developing interactive processes to create experiences, and measuring customer responses using appropriate performance metrics [15]. While CXM is similar to customer relationship management (CRM) in its use of data analytics to understand customer preferences, it differs in that CRM focuses primarily on improving sales and operational processes, whereas CXM emphasizes creating holistic and emotional experiences across all touchpoints [11, 21, 33]. Therefore, while CRM aims to create value for firms, CXM seeks to generate value for customers by delivering unique and positive experiences [18, 21].

Leveraging contemporary technologies is crucial for managing the customer experience, as it has become almost impossible to do so effectively across the myriad touchpoints that characterize today's market environment. In omnichannel environments, it has become imperative to integrate information technology, service operations, logistics, human resources, and external partners [21]. Artificial intelligence is being adapted to marketing strategies, product management, pricing, place management, and promotion management—in other words, to every component of the marketing mix—and gaining insights using large volumes of customer and transactional data, including text, voice, image, and facial expression data [8, 25, 34]. Decision-making, customer segmentation, customer service, tracking touchpoints along the customer journey, positioning, personalized recommendations, content creation, pricing and placement management, experiential marketing, branding, sales forecasting, and market research are among the areas where AI has become deeply embedded in the marketing field [2, 17]. More recently, as the use of AI tools has expanded, the concept of generative AI has emerged, enabling artificial intelligence to produce marketing content such as text, audio, and images similar to that created by human marketing professionals, representing a groundbreaking development for businesses [16]. Supported by data analytics, AI technologies enhance

consumer experiences by enabling businesses to learn about consumer preferences and shopping patterns. Moreover, strategically deploying these technologies at key customer touchpoints can provide firms with significant competitive advantages [3]. Personalization is essential for delivering an enhanced customer experience [16, 20], and AI solutions provide consumers with personalized experiences beyond human capability.

In marketing and CXM activities, different types of AI are inevitably utilized in strategic stages such as research, strategy development, and implementation. Marketers can benefit from mechanical, thinking, and feeling AI technologies, which are categorized according to AI's ability to perform tasks of increasing cognitive and emotional complexity [12]. Mechanical AI is used to automate repetitive and routine tasks, while thinking AI is designed to process data to reach new conclusions or decisions. Feeling AI is used to analyze two-way interactions involving human emotions and affective responses [12]. For example, all three types of AI can be used to understand the potential impact of a new technological product planned for market launch, to create an appropriate strategy, and to implement it. Mechanical AI can scrape data from platforms such as social media and e-commerce sites and perform market segmentation based on the data collected. Thinking AI can categorize consumer expectations into themes such as design, performance, technological criteria, and user experience using natural language processing, and offer product recommendations to customers based on these criteria. Feeling AI can understand the emotional tone of data collected by mechanical AI, such as customer comments and reviews, and create marketing content based on this emotional tone. In the conceptual framework developed in this study, which will be discussed in detail later, AI tools are considered useful at all levels of CXM, so they are placed in a horizontal cross section in our conceptual framework (Fig. 1).

CXM and AI are often discussed in the context of big data analytics. This is natural because big data analytics is recognized as the raw fuel for AI technology [17]. Big data analytics involves the collection, organization, and utilization of large-scale data, supporting managers in developing insights into consumer preferences and behaviors [18]. Holmlund et al. [11] proposed a strategic framework that integrates CXM and big data analytics. According to this strategic framework, different structured or unstructured, requested, or unsolicited

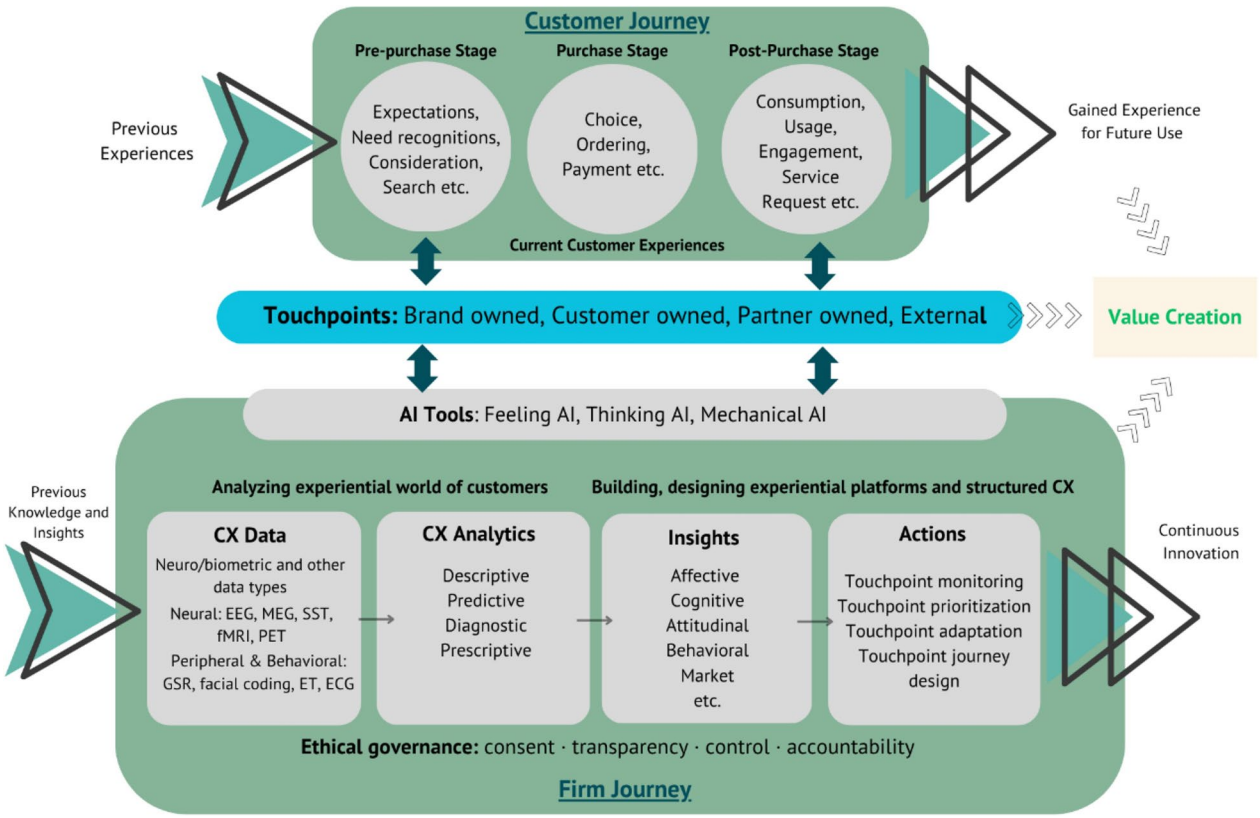


Fig. 1 Conceptual framework for CXM in the context of AI and neuromarketing

customer data are processed through analytical processes such as descriptive, predictive, diagnostic, and prescriptive according to the needs, and meaningful customer information is generated for businesses. This information generates valuable insights into market dynamics, various touchpoints, and customers' attitudes and behaviors.

In the customer journey, it is important for CXM to understand where customers are positioned, which variables and touchpoints influence them the most, and the factors that lead them to continue or discontinue their journey. Understanding the customer, who is the subject of this journey, has always been the leading subject of marketing and many methods have been developed to understand the customer. These methods include traditional research approaches, big data analytics techniques that extract information from the vast amount of data left behind by customers, and as discussed later, neuroscientific methods that aim to generate insights into consumers' cognitive and physiological responses. In particular, since customer experiences are triggered by specific stimuli [6] and these triggers influence customer attitudes and behaviors over time, it is important to understand how consumers respond to different stimuli, and one of the most effective ways to do so is through the use of neuroscience tools.

In neuroscience, neural processes that occur during behavior are measured in real time through methods that directly assess neural activity (electromagnetic and metabolic, such as EEG and fMRI), methods that record peripheral physiological, facial, or behavioral/attentional activity (such as GSR, facial coding, and eye-tracking), and methods for manipulating neural activity [22]. The use of neuromarketing data, which enables the generation of valuable insights into sophisticated and complex human behavior, offers significant opportunities for customer experience management. Neuromarketing and AI have a mutually beneficial, bidirectional relationship. Specifically, neuromarketing generates highly complex biometric and neural data from consumers' brains and bodies (e.g., EEG, eye-tracking, GSR, facial coding). On their own, these datasets are vast and unstructured, making it difficult to derive meaningful insights. At this point, artificial intelligence and machine learning play a crucial role. AI processes these data and translates them into faster, more accurate, and predictive analytical outputs [10]. Conversely, the most valuable input for AI algorithms is large-scale, rich data grounded in real human behavior. Neuromarketing provides biometric traces of consumers' emotions, attention levels, and preferences, thereby facilitating AI's "learning" process. As a result, AI can perform more precise segmentation, develop personalized campaigns, and deliver more accurate recommendations to target audiences. A productive balance between neuroscience, artificial intelligence, and marketing can provide marketers with opportunities to discover and strategize for niche markets. Neuromarketing data from diverse neuroscientific sources can be integrated with AI to create valuable applications [36]. The application of neuromarketing data to AI will expand the boundaries of neuromarketing, and AI systems fed with such data will advance efforts to understand the consumer to the next level. Personalization is a key factor for CXM [16, 20], and neuromarketing data provide a valuable input for AI to deliver personalized experiences.

Studies that bridge AI and neuroscience in the literature generally focus on the imitation of the human brain by AI [5]. Studies on AI in marketing have generally focused on big data analytics and their application to business contexts [11, 27, 30]. However, in contrast to the academic literature, companies have implemented methods to enhance customer experience by integrating neuromarketing data into AI systems. In addition, many information technology companies collect neuromarketing data, train AI systems with these data, and market such systems—for instance, Neuro-Insight, Emotiv, NeuroFocus, Innerscope Research, and Ipsos. Moreover, it is widely recognized that corporations such as Procter & Gamble, Coca-Cola, Nestlé, General Motors, Campbell, PayPal, Walmart, and IKEA have been actively involved in neuromarketing research and in the adoption of artificial intelligence and machine learning technologies [10].

Framework development and illustrative vignettes

In developing the conceptual framework for this study, we drew upon Lemon and Verhoef's [21] study on customer journey stages and on Schmitt's [29] and Holmlund et al.'s [11] works on the CXM process. These models were subsequently expanded and positioned within a neuromarketing framework based on illustrative secondary vignettes. As a result of the literature review and the mapping of these vignettes onto the framework, the conceptual framework presented in Fig. 1 was developed. This conceptual framework focuses on the neuromarketing-to-AI direction of this reciprocal relationship. To illustrate the proposed framework, three enterprises were examined: Liberty Group SA (insurance sector, South Africa), TUI Group (tourism and travel sector, Germany), and Zyro (software and technology sector, Lithuania). These companies were selected for their sectoral diversity, for the integration of AI and neuromarketing tools in customer experience management, and for the public availability of data. The vignettes illustrate how AI–neuromarketing collaboration has been described across different contexts—Liberty through an AI-based chatbot supported by neuromarketing analytics, TUI through emotion recognition for travel personalization,

and Zyro through AI-generated heatmaps informed by eye-tracking data—and are used to exemplify the categories of the proposed framework rather than to test it.

It is appropriate to examine this conceptual framework from two perspectives. The first dimension is the customer journey, which begins with the customer's recognition of a need and continues through their behavior in post-purchase processes. This stage represents the classic consumer decision-making process. It consists of pre-purchase, purchase, and post-purchase stages, each influenced by numerous stimuli and elements that affect consumer decisions. There are many marketing research studies on customer intentions, behaviors, and perceptions. In this article, we are concerned with the management of stimuli at the points where firms come into contact with customers. Touchpoints are communication environments where customers and firms intersect. These may include physical stores or advertisements encountered while browsing social media. Touchpoints can be owned by the customer, the firm, the firm's partners, or the customer's social and external environment. Each of these touchpoints is regarded as a communication channel, and businesses must deliver consistent messages across them to enhance customer experience.

The second component is what we call the firm journey. If the customer journey is defined as the process beginning with the search to satisfy a need and ending with the post-purchase evaluation of a product, the firm's CXM process can be regarded as the firm's journey—from employing neuroscientific methods to identify needs, to developing corresponding solutions, and ultimately to creating value. Firms start this journey with the insights and knowledge they already possess. To progress further, they require customer data. At this stage, the use of neuroscientific methods that measure neural processes occurring during behavior—whether through direct neural measures or peripheral/biometric and behavioral measures—represents an essential first step for CXM to achieve a deep understanding of customers and deliver optimal outcomes that meet their needs.

Customer data obtained in this way are processed through data analytics techniques and transformed into actionable information. This information enables the generation of insights into customers—from the moment they encounter a stimulus, to their use of a product, to observing others consuming a product of interest. The critical issue is whether the insight gained can be realized at a touchpoint where the firm can intervene and whether a strategy can be formulated accordingly. Because customers are influenced by numerous variables across many touchpoints, CXM aims to ensure a consistent experience at nearly every one of them. To maintain such consistency, AI tools provide substantial advantages.

As illustrated in Fig. 1, firms create, design, or configure experiential platforms to enhance customer experience and generate value based on the insights they acquire. The strategies derived from these insights are generally categorized as touchpoint monitoring, touchpoint prioritization, touchpoint adaptation, and touchpoint-journey design. Insights may reveal the need to monitor customer behavior across different touchpoints and other interacting variables. Some touchpoints may exert greater influence than others, or firms may need to develop entirely new designs that transform customer experience at a particular point of contact. It may even become evident that a new touchpoint must be created.

Indeed, these touchpoint alternatives are broad enough to encompass all marketing strategies. As discussed in the theoretical section, CXM is an overarching concept that integrates numerous customer-related constructs, including customer satisfaction, perception, motivation, loyalty, engagement, relationship marketing, and customer centricity. The framework in Fig. 1 serves as a comprehensive, bird's-eye-view map encompassing all these dimensions. What must be emphasized is that the individual customer strategies implemented at each touchpoint need to remain coherent and integrated within the broader picture—a task that is highly challenging for firms to achieve. If they succeed, the ultimate outcome is value. Leveraging the possibilities offered by today's technologies is therefore vital to accomplishing this complex endeavor.

This study is a conceptual article that uses illustrative secondary vignettes to map the proposed framework, focusing on CXM, AI, and neuromarketing. This approach draws inspiration from the case research tradition [9, 35], which recommends grounding conceptual development in real-world settings, particularly where prior conceptualization is limited and where the field is changing rapidly [4]. However, the cases examined in this study are derived from publicly available secondary sources—company websites, press releases, and industry coverage—rather than from primary data collected by the authors within organizational settings. They are therefore used strictly as conceptual illustrations of the proposed framework and should not be interpreted as case studies in the methodological sense described by Eisenhardt [9] and Voss et al. [35]. To make this approach transparent, the vignettes were selected using three criteria: sectoral diversity, integration of AI and neuromarketing tools, and the public availability of sufficiently detailed information. Information was drawn from company websites and press coverage. Because the underlying information originates from company and public sources rather than independently verified data, the inferences that can be drawn from the vignettes are necessarily limited:

They can illustrate how the framework's categories map onto real-world practice, but they cannot establish the accuracy of the reported outcomes, isolate the causal effect of AI–neuromarketing integration, or support generalization beyond the cases discussed. The cases are presented as illustrative vignettes that show how the proposed framework can be mapped onto real-world practices, rather than as exhaustive empirical validation of all framework dimensions. The companies chosen for analysis are sector pioneers that integrate artificial intelligence and neuromarketing collaborations, possess strong technological infrastructures, and demonstrate openness to innovation. Considering that these practices are still emerging within the sector, the number of firms implementing them remains limited. Both the desired depth of observation and the restricted number of applicable cases guided the decision to proceed with three cases. The scope of the study is confined to B2C firms,the rationale for this scope decision is set out in the conclusion. As a result of the illustrative vignettes discussed in the subsequent sections, Table 1 is generated.

The primary contribution of this framework lies in positioning neuro/biometric signals as a distinct class of data that can inform customer experience management decisions beyond behavioral and transactional data. The framework enables these signals to be transformed, through AI-enabled analytical processes, into inferences regarding customers' affective, cognitive, and attentional states, and links these inferences to concrete touchpoint decisions such as monitoring, prioritizing, adapting, and designing. In this way, the proposed approach goes beyond CXM models based solely on behavioral data by providing a theoretical and practical basis for more nuanced managerial interventions, including the identification of affective friction throughout the customer journey, the analysis of visual attention allocation, and the matching of content to customers' emotional states. Lemon and Verhoef [21] and Holmlund et al. [11] conceptualize the customer journey and CXM analytics primarily around behavioral and transactional data,Huang and Rust [12] classify AI capabilities (mechanical, thinking, feeling) but do not specify the data inputs that feed each type. The present framework's contribution is to identify neuro/biometric data (EEG, GSR, facial coding, eye-tracking) as a distinct data class that gives AI-driven analytics access to customers' affective, attentional, and cognitive states not directly observable in transactional or behavioral records. Two mechanisms follow from this: First, neuro/biometric signals allow AI systems to infer affective and attentional states in situ, at the moment of the customer–firm interaction, rather than retrospectively from purchase or clickstream data,second, these inferences can be mapped onto specific managerial actions—touchpoint monitoring, prioritization, adaptation, and journey design—creating a more direct link between psychophysiological signals and firm-level decisions than existing behavioral-data frameworks provide. Positioning the contribution around these two mechanisms, rather than around the case narratives themselves, clarifies that the framework's novelty lies in the data source and its mapping onto managerial decisions, building on Lemon and Verhoef [21] and Holmlund et al. [11].

From an ethical perspective, the framework should be applied with careful attention to informed consent, transparency, data minimization, consumer choice, and algorithmic accountability. Consumers should understand how their neurophysiological or biometric data are collected and used, and they should be able to opt out without losing access to essential services. Firms should also explain, as far as possible, how AI systems use these data to make inferences about customers. Appropriate safeguards are needed to reduce the risk that personalized

Table 1 Mapping the vignettes onto the framework

Illustrative Vignettes	Customer journey	AI tools	Touch points	Firm journey				Value creation mechanism	Value creation ends up with:
	Customer journey stage			Neuro/ biometric data	CX analytics	Insights	Actions		
Vignette 1: Liberty	Purchase stage	Feeling AI	Brand owned	Neural: EEG Peripheral & behavioral: GSR, FACS, ET	Predictive	Affective and cognitive	Touchpoint monitoring	Optimization of the user interface	Time efficiency and purchasing behavior
Vignette 2: TUI	Pre-purchase stage	Feeling AI	Brand owned	Peripheral & behavioral: FACS	Predictive	Affective	Touchpoint adaptation	Advanced suggestion system	Arouse curiosity in the customer
Vignette 3: Zyro	Pre-purchase stage	Thinking AI	Partner owned	Peripheral & behavioral: ET	Predictive	Attentional	Touchpoint-journey design	Platform optimization	Increased response to stimuli

practices may exploit customers' emotional or cognitive responses. These principles should be considered throughout the firm journey presented in Fig. 1, from data collection and analysis to insight generation and action.

Discussion

In this study, the proposed conceptual framework illustrates, in a step-by-step manner, how CXM can be redefined, addressed, and implemented in the context of AI and neuromarketing. The reciprocal interaction between the customer journey (pre-purchase, purchase, and post-purchase) and the firm journey (data collection, analytics, insights, and actions) is explained through different touchpoints and AI tools.

Figure 1 illustrates how neural (e.g., EEG, fMRI) and peripheral/biometric (e.g., GSR, ET, facial coding) data can be processed with AI-supported analytics (descriptive, predictive, diagnostic, and prescriptive) and transformed into emotional, cognitive, and behavioral insights about consumers. The framework suggests that these insights may help firms optimize the customer journey through monitoring, prioritization, and adaptation of touchpoints. In this way, the framework suggests that value creation can be achieved both on the customer side (enhanced experience, personalization, satisfaction) and on the firm side (continuous innovation). Table 1 shows how the proposed framework is operationalized through the illustrative vignettes; as noted above, the reported outcomes are drawn from company and public sources and are not independently verified. In the Liberty case, a combination of neural data (EEG) and peripheral physiological data (GSR, facial coding, eye-tracking) was reportedly used to inform emotional and cognitive responses, and user interface optimization was reportedly achieved through touchpoint monitoring, with the firm reporting gains in time and cost efficiency and in purchasing behavior. In the TUI case, peripheral data such as facial coding (FACS) were reportedly employed to infer consumers' affective reactions—and, more tentatively and with less evidentiary support, cognitive responses—while advanced recommendation systems were reportedly implemented through the adaptation of touchpoints, with the firm associating this with outcomes such as stimulating customer curiosity. In the Zyro case, a predictive attention model together with AI-supported touchpoint journey design were reportedly applied, which the provider associates with increased responsiveness of consumers to stimuli. Taken together, the framework proposes that when neuromarketing and AI are integrated into customer experience management, the diversity of data—including both neural and peripheral/biometric signals—may provide firms with insights into customers that transactional and behavioral records alone do not capture. Within the framework, AI-based analytical processes translate these data into predictive and strategic insights, which in turn inform concrete actions such as user interface optimization, recommendation systems, and platform design. The framework posits that these actions are the mechanism through which value may be created within the customer experience, on both the customer side (satisfaction) and the firm side (efficiency and competitive advantage). The vignettes are consistent with this logic but do not establish it: They are illustrative observations drawn from company and public sources rather than causal evidence, and testing the proposed relationships would require primary empirical research.

Compared to previous studies, the proposed framework both aligns with and extends earlier findings. For example, similar to Holmlund et al. [11] and Kumar et al. [17], our framework is consistent with the view that big data analytics play a fundamental role in CXM. The proposed framework goes further by suggesting that, beyond big data, neuromarketing may offer unique consumer-level insights that AI can learn from, thereby broadening the scope of customer understanding. While Wedel and Kannan [36] emphasized the potential of neuroscience for CX research and noted the lack of empirical applications, the vignettes indicate that firms such as Liberty, TUI, and Zyro publicly describe initiatives of this kind, suggesting that such applications are beginning to appear in practice. In this sense, our study offers a conceptual response to the unanswered calls for research on the role of neuroscientific methods in CXM [21, 36], which future empirical work could test directly. Earlier literature on AI in marketing has largely emphasized personalization based on behavioral or transactional data [2, 8]. In contrast, our study suggests that personalization can be deepened through emotional and cognitive insights derived from neuromarketing, offering a richer form of customer engagement. This aligns with the growing recognition that AI-enabled customer experience extends beyond efficiency gains to encompass human-like engagement, authenticity, and responsible implementation [28]. More broadly, these developments underscore that CXM itself is increasingly understood as a strategically differentiated process rather than a uniform practice [38]. Furthermore, while Lim [22] described the methodological potential of neuroscientific tools in understanding consumer behavior, our vignettes illustrate how these tools may be practically embedded into AI-driven systems, pointing to a possible route for overcoming earlier limitations of scalability and application. Furthermore, while it is evident that applying neuromarketing techniques and artificial intelligence in marketing raises

certain ethical concerns, it is equally important to recognize that similar issues arise in the context of CXM, necessitating preventive measures. Goncalves et al. [10] emphasize that the use of AI and neuromarketing brings forth fundamental ethical challenges, including a lack of transparency, manipulation, protection of human subjects, racial bias, and data security.

Taken together, these insights suggest that the convergence of AI and neuromarketing advances CXM beyond traditional big data analytics toward a more nuanced, emotion-based understanding of customers. By illustrating practical applications across three industries, our study highlights how this integration may enable firms not only to monitor and adapt touchpoints but also to design entirely new customer journeys. Thus, while our findings are consistent with earlier calls to incorporate new data and analytics into the analysis of customer experience [21], they also provide concrete illustrations of how such integration can be achieved, offering both theoretical and practical contributions to the field.

Conclusions

The rapid development of artificial intelligence is transforming marketing practices and competitive dynamics. In today's business environment, it is crucial for companies to invest in AI technologies to remain competitive. At the same time, they must explore new opportunities to identify potential growth areas and remain in the blue ocean.

This study conceptualizes the integration of neuromarketing and AI within the context of CXM through a conceptual framework and outlines the potentially transformative role of this integration. It argues that the convergence of neuromarketing with AI and machine learning may offer firms more granular insights, stronger predictive capabilities, and more finely personalized interaction opportunities. While previous research has predominantly portrayed AI as a tool for big data analytics and neuromarketing as a method for understanding neural mechanisms, publicly available accounts of company practice suggest that some firms are beginning to use neuromarketing data as an input to AI systems in a CXM context. By developing a conceptual framework, this study seeks to bridge the gap between theory and practice and to underscore the strategic significance of neuromarketing–AI integration for both marketing literature and practice.

From a managerial perspective, the framework points to the potential of these technologies to transform B2C strategies by enabling firms to segment consumers more effectively, design personalized campaigns, and optimize customer journeys. However, the study also emphasizes that collaboration between AI and neuromarketing gives rise to ethical dilemmas that cannot be overlooked. While these technologies provide firms with unprecedented opportunities to deliver personalized and seamless experiences, they simultaneously raise concerns about consumer manipulation, data privacy, and human rights. Therefore, companies adopting such approaches must ensure transparency, obtain informed consent, and establish clear ethical standards to safeguard consumer trust.

This study has several limitations that follow directly from its design. First, it is a conceptual article: The proposed framework is not empirically tested, and no claims are made about the size or direction of any performance effects. Second, the three vignettes are drawn from publicly available secondary sources—company websites and press releases—some of which are promotional in nature. The outcomes reported by the firms have not been independently verified, and the vignettes therefore illustrate how the framework's categories map onto described practice rather than demonstrating that the practices produce the stated results. Third, with three purposively selected firms, the vignettes do not support statistical or theoretical generalization, and no causal inference can be drawn about the effect of AI–neuromarketing integration on customer or firm outcomes.

The study is also bounded by a deliberate scope decision. We focus on B2C contexts because the publicly documented applications of AI–neuromarketing integration are currently consumer-facing; this reflects the availability of observable practice rather than any inherent limitation of neuromarketing or CXM, both of which are equally relevant to organizational buyers. Future research could examine how the framework travels to B2B settings, where buying centers, longer sales cycles, professional risk perceptions, and relational dynamics may alter the role of neuro/biometric data. Empirical work—experimental studies, primary multiple-case research with organizational access, or field tests of specific touchpoint interventions—would allow the framework's propositions to be tested directly. Finally, the ethical issues raised here (data management, consumer control, and the responsible use of neuroscientific findings), along with the role of regulatory and governance mechanisms, should remain a priority for both academic research and practical applications.

Appendix 1

Vignette 1: An Insurance Company — Liberty Group SA
Liberty is an insurance company based in South Africa that previously received short-term insurance applications through its call centre. To reduce costs and enhance the customer experience, the company developed an AI-based chatbot, deployed on its website and messaging channels, with a live-chat handover to a call-centre agent where a customer becomes stuck [24]
To understand how users interacted with the chatbot interface, Liberty engaged the neuromarketing consultancy Neural Sense, which was tasked with optimising the chatbot's development through iterative consumer neuroscience testing. The stated objectives were to establish what the chatbot should look like, what personality it should possess and the tone and register it would use in communicating with clients, and to optimise the end-to-end user journey so that it would be smooth, effortless and intuitive (Neural Sense [26])
While customers interacted with the chatbot, the consultancy reports using eye-tracking to determine what attracts and holds attention, galvanic skin response and facial coding to measure the type and degree of affective response, and electroencephalography to assess cognitive flow, motivation and engagement across the user journey (Neural Sense [26],iMotions [14]). The resulting data were used to establish how the interaction between the chatbot and the customer could be optimised without human intervention, and informed decisions on the chatbot's tasks, tone of voice and visual representation
At a pre-launch media briefing, the chief executive of Liberty Short-term Insurance stated that the intended quote experience was a quote in five minutes and a policy purchase in eight minutes, against a call-centre benchmark of thirty to forty-five minutes for a quote [24]. *These are objectives announced before the chatbot's full commercial release and reported in trade press; they are not independently verified performance measures, and the source does not attribute them to the neuromarketing work*

Vignette 2: A travel retail company: TUI
TUI, one of Europe's largest tourism companies, partnered with Realeyes, a company specialising in facial coding and emotion measurement, to make holiday selection more engaging. Together they developed the "Destination U" prototype, unveiled in October 2017 and presented by TUI as a first of its kind in travel retail (TUI [32])
The prototype presents a rapid series of evocative moving images of destinations and experiences while a camera records the customer's facial responses. An algorithm computes each subtle facial response and uses the resulting data to calculate a "perfect holiday" prescription, with the images shown varying according to the customer's reactions (TUI [32],Travel Weekly [31]). The technology is described as using sophisticated facial coding and emotion measurement to tap into responses that are largely subconscious, with the stated aim of helping customers unlock travel possibilities they might not otherwise have considered (TUI [32])
At the time of the announcement the system was described as a prototype still undergoing testing rather than a fully deployed service (TUI [32]). *No source consulted discloses the composition of the dataset on which the underlying emotion models were trained*

Vignette 3: A software company: Zyro
Zyro is a Lithuania-based software company headquartered in Kaunas, founded in 2019 and operating under Hostinger. It offers website building, e-commerce and AI-based digital tools that enable users to design professional websites without coding skills. Zyro's tools have since been consolidated into Hostinger Website Builder; this vignette describes the service as documented at the time of data collection
One of its key AI features is the AI Heatmap. The provider describes it as a tool that gives insights into what website visitors will focus on before they actually visit, so that decisions can be made before money is spent on user acquisition [39]. The tool belongs to the class of visual-attention (saliency) models, which are conventionally trained on gaze data in order to predict where viewers will look. Independent review coverage describes it as an AI-enabled heatmap showing where a page's viewers are likely to focus, overlaid on the site design to highlight the content that will stand out when users land on the page or scroll down it (IT Pro [13])
These AI-generated heatmaps help website owners understand likely user attention, optimise site layout, and place key elements such as calls to action where users are most likely to notice them. *The provider does not disclose the composition of the training data underlying the model. In terms of the proposed framework, attention-based data therefore enter the firm journey through a pre-trained predictive model rather than through primary data collection by the focal firm*

Appendix 2
Sources used for the illustrative vignettes

Vignette	Source details	Type and status
Liberty	Mzekandaba, S. [24] 'Liberty readies for insurance chatbot'. ITWeb, 15 March	Trade press—independent
Liberty	Neural Sense [26] 'The future of neuromarketing in a developing world'. The Science of Persuasion, 7 August	Consultancy blog—promotional
Liberty	iMotions [14] 'How to conduct successful neuromarketing studies' [Neural Sense interview]	Vendor blog—semi-independent
TUI	TUI UK [32] 'World's leading travel company reveals the future of holiday shopping…'. TUI UK Media Centre, 18 October	Corporate press release—promotional
TUI	Travel Weekly [31] 'Tui unveils facial recognition technology that predicts customers' ideal holiday'. 18 October	Trade press—independent
Zyro	Zyro [39] 'Zyro steps closer to changing the way websites are made'. PR Newswire, 28 January	Corporate press release -promotional
Zyro	IT Pro [13] 'Zyro website builder review'. 5 October	Independent product review—independent

Full details and access dates are given in the Appendix References. Publisher independence does not imply independent verification of corporate claims: the Liberty timing figures are pre-launch targets, and none of the sources discloses the training data behind Destination U or the Zyro AI Heatmap

Abbreviations

AI	Artificial intelligence
B2B	Business-to-business
B2C	Business-to-consumer
CRM	Customer relationship management
CXM	Customer experience management
EEG	Electroencephalography
ET	Eye tracking
fMRI	Functional magnetic resonance imaging
GSR	Galvanic skin response

Author contributions

B.T. and A.H. contributed to the conceptualization, investigation, and methodology of the study. A.H. supervised the research process. B.T. prepared the original draft of the manuscript. Both authors contributed to the review and editing of the manuscript. All authors read and approved the final version.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

In this article, a large language model was used only for linguistic support, specifically to assist with English phrasing and clarity. All suggestions were subsequently reviewed and verified by the authors.

Consent for publication

Not applicable.

Competing interests

Not applicable.

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References

1. Alsharif AH, Wang J, Isa SM et al (2025) The synergy of neuromarketing and artificial intelligence: A comprehensive literature review in the last decade. Future Business Journal 11:170. https://doi.org/10.1186/s43093-025-00591
2. Ameen N, Sharma GD, Tarba S, Rao A, Chopra R (2022) Toward advancing theory on creativity in marketing and artificial intelligence. Psychol Mark 39(9):1802–1825. https://doi.org/10.1002/mar.21699
3. Ameen N, Tarhini A, Reppel A, Anand A (2021) Customer experiences in the age of artificial intelligence. Comput Hum Behav 114:106548. https://doi.org/10.1016/j.chb.2020.106548
4. Amit R, Zott C (2001) Value creation in E-business. Strateg Manag J 22(6–7):493–520. https://doi.org/10.1002/smj.187
5. Bermudez-Contreras E, Clark BJ, Wilber A (2020) The neuroscience of spatial navigation and the relationship to artificial intelligence. Frontiers in Computational Neuroscience 14:63.https://doi.org/10.3389/fncom.2020.00063
6. Brakus JJ, Schmitt BH, Zarantonello L (2009) Brand Experience: What is It? How is it Measured? Does it Affect Loyalty? Journal of Marketing. https://doi.org/10.1509/jmkg.73.3.052
7. Bullemore-Campbell J, Díaz Tautiva JA (2025) Integrating neuroscience into B2B sales research: foundations, frontiers, and expert directions. Manag Rev Q. https://doi.org/10.1007/s11301-025-00550-4
8. Davenport T, Guha A, Grewal D, Bressgott T (2020) How artificial intelligence will change the future of marketing. J Acad Mark Sci 48(1):24–42. https://doi.org/10.1007/s11747-019-00696-0
9. Eisenhardt KM (1989) Building theories from case study research. Acad Manag Rev 14(4):532. https://doi.org/10.2307/258557
10. Goncalves M, Hu Y, Aliagas I, Cerdá LM (2024) Neuromarketing algorithms' consumer privacy and ethical considerations: challenges and opportunities. Cogent Bus Manag 11(1):2333063. https://doi.org/10.1080/23311975.2024.2333063
11. Holmlund M, Van Vaerenbergh Y, Ciuchita R, Ravald A, Sarantopoulos P, Ordenes FV, Zaki M (2020) Customer experience management in the age of big data analytics: a strategic framework. J Bus Res 116:356–365. https://doi.org/10.1016/j.jbusres.2020.01.022
12. Huang M-H, Rust RT (2021) A strategic framework for artificial intelligence in marketing. J Acad Mark Sci 49:30–50. https://doi.org/10.1007/s11747-020-00749-0
13. IT Pro (2022) 'Zyro website builder review', 5 October. Accessed 26.07.2026. https://www.itpro.com/development/web-development/368219/zyro-website-builder-review
14. iMotions (n.d.) 'How to conduct successful neuromarketing studies' [Neural Sense interview]. Accessed 26.07.2026. https://imotions.com/blog/learning/best-practice/successful-neuromarketing-studies/
15. Jain R, Aagja J, Bagdare S (2017) Customer experience – a review and research agenda. JSTP 27(3):642–662. https://doi.org/10.1108/JSTP-03-2015-0064
16. Kshetri N, Dwivedi YK, Davenport TH, Panteli N (2024) Generative artificial intelligence in marketing: applications, opportunities, challenges, and research agenda. Int J Inf Manage 75:102716. https://doi.org/10.1016/j.ijinfomgt.2023.102716
17. Kumar V, Ramachandran D, Kumar B (2021) Influence of new-age technologies on marketing: a research agenda. J Bus Res 125:864–877. https://doi.org/10.1016/j.jbusres.2020.01.007
18. Ledro C, Nosella A, Vinelli A (2022) Artificial intelligence in customer relationship management: literature review and future research directions. JBIM 37(13):48–63. https://doi.org/10.1108/JBIM-07-2021-0332
19. Ledro C, Nosella A, Vinelli A, Dalla Pozza I, Souverain T (2025) Artificial intelligence in customer relationship management: a systematic framework for a successful integration. J Bus Res 199:115531. https://doi.org/10.1016/j.jbusres.2025.115531
20. Lemke F, Clark M, Wilson H (2011) Customer experience quality: an exploration in business and consumer contexts using repertory grid technique. J Acad Mark Sci 39(6):846–869. https://doi.org/10.1007/s11747-010-0219-0
21. Lemon KN, Verhoef PC (2016) Understanding customer experience throughout the customer journey. J Mark 80(6):69–96. https://doi.org/10.1509/jm.15.0420
22. Lim WM (2018) Demystifying neuromarketing. J Bus Res 91:205–220. https://doi.org/10.1016/j.jbusres.2018.05.036
23. Lipkin M, Heinonen K (2022) Customer ecosystems: exploring how ecosystem actors shape customer experience. J Serv Mark 36(9):1–17. https://doi.org/10.1108/JSM-03-2021-0080
24. Mzekandaba, S. (2018) 'Liberty readies for insurance chatbot', ITWeb, 15 March. Accessed 26.07.2026. https://www.itweb.co.za/article/liberty-readies-for-insurance-chatbot/Olx4z7kgbg8756km
25. Nalbant KG, Aydin S (2025) A bibliometric approach to the evolution of artificial intelligence in digital marketing. Int Mark Rev 42(2/3):179–218. https://doi.org/10.1108/IMR-04-2024-0132
26. Neural Sense (2018) 'The future of neuromarketing in a developing world', The Science of Persuasion, 7 August. Accessed 26.07.2026. https://www.thescienceofpersuasion.com/single-post/2018/08/07/future-of-neuromarketing-in-a-developing-world
27. Paschen J, Kietzmann J, Kietzmann TC (2019) Artificial intelligence (AI) and its implications for market knowledge in B2B marketing. J Bus Ind Mark 34:1410–1419. https://doi.org/10.1108/JBIM-10-2018-0295
28. Sahut J, Laroche M (2025) Using artificial intelligence (AI) to enhance customer experience and to develop strategic marketing: an integrative synthesis. Comput Hum Behav 170:108684. https://doi.org/10.1016/j.chb.2025.108684

29. Schmitt BH (2003) Customer experience management: a revolutionary approach to connecting with your customers. The Free Press, New York
30. Tao S, Kim H-S (2022) Online customer reviews: insights from the coffee shops industry and the moderating effect of business types. TR 77(5):1349–1364. https://doi.org/10.1108/TR-12-2021-0539
31. Travel Weekly (2017) 'Tui unveils facial recognition technology that predicts customers' ideal holiday', 18 October. Accessed 26.07.2026. https://travelweekly.co.uk/news/tui-unveils-facial-recognition-technology-that-predicts-customers-ideal-holiday
32. TUI UK (2017) 'World's leading travel company reveals the future of holiday shopping with cutting edge prototype that taps into the subconscious', TUI UK Media Centre, 18 October. Accessed 26.07.2026. https://www.tui.co.uk/press/destination-u/
33. Verhoef PC, Lemon KN, Parasuraman A, Roggeveen A, Tsiros M, Schlesinger LA (2009) Customer experience creation: determinants, dynamics and management strategies. J Retail 85(1):31–41. https://doi.org/10.1016/j.jretai.2008.11.001
34. Verma S, Sharma R, Deb S, Maitra D (2021) Artificial intelligence in marketing: systematic review and future research direction. Int J Inf Manag Data Insights 1(1):100002. https://doi.org/10.1016/j.jjimei.2020.100002
35. Voss C, Tsikriktsis N, Frohlich M (2002) Case research in operations management. Int J Oper Prod Manag 22(2):195–219. https://doi.org/10.1108/01443570210414329
36. Wedel M, Kannan PK (2016) Marketing analytics for data-rich environments. J Mark 80(6):97–121. https://doi.org/10.1509/jm.15.0413
37. Wirth N (2018) Hello marketing, what can artificial intelligence help you with? Int J Mark Res 60(5):435–438. https://doi.org/10.1177/1470785318776841
38. Wirtz J, Kowalkowski C, Jaakkola E, Holmlund M, Ulaga W, Ahmed T (2025) Customer experience management in B2B markets: CXM value propositions and archetypical CXM strategies. J Bus Res 189:115165. https://doi.org/10.1016/j.jbusres.2024.115165
39. Zyro (2020) 'Zyro steps closer to changing the way websites are made', PR Newswire, 28 January. Accessed 26.07.2026. https://www.prnewswire.com/in/news-releases/zyro-steps-closer-to-changing-the-way-websites-are-made-866982911.html

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